

6 Mar 2023
Four Seasons Hotel
Four Seasons Hotel Branch"
138 Chifeng Road, Heping District,
Tianjin 300041, China

For the attention of Mr. Leon Leo Li
Director of Engineering

Subject to contract

Ref MW/FS/TJ/EA/1

Dear Sir,

Proposal for Energy Audit Revisit Four Seasons Hotel Tianjin

Further to our discussion, we are pleased to detail our proposed Energy Assessment service for the above facility per standard Four Seasons APAC zone agreement. We will update previous findings and report current status as well as setting up the Four Seasons APAC- wide energy savings tracking system - Action Management Interface (AMI) details. Details and terms etc. are indicated below.

SCOPE:

1. To update previous investigation findings for energy and utilities usage and cost data from available direct invoices, power house efficiency and to assess apportionment / recommend supplementary sub-metering where appropriate.
2. To review outcome of previous recommendations for efficient operating strategies for equipment and update for the installation in its present form.
3. To review progress with previous energy saving opportunities, identify fresh improvement options. Rank energy saving opportunities for retrofit energy saving solutions, including consideration of cost, savings potential, ease and practicality of implementation. This would include but not necessarily be confined to:
 - a. Co-generation
 - b. PV electricity and water turbines
 - c. Lighting
 - d. Hot water distribution



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System
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- e. Water extraction, purification, recycling and economy measures
- f. Waste water treatment
- g. Kitchens and refrigeration
- h. Laundry equipment
- i. Steam and Hot water boilers
- j. Pools, spa and water features
- k. Refrigeration heat recovery
- l. MVAC including central and distributed systems considering plant efficiency and point of use controls
- m. Metering and sub-metering
- n. Boat operations Diesel and gasoline usage.

DELIVERABLES

We will allocate 3-4 days site investigation from our executive/ senior project engineer + 2 project engineers for each site to conduct a comprehensive assessment of issues above and will report via: -

- 1) Narrative and graphical descriptions of existing usage patterns, updated and fresh savings opportunities and inclusion of related photos to amplify understanding of the locations and concepts for savings per scope above
- 2) Best estimate calculations for savings and investment costs, presented in a logical tabular form
- 3) Tabulated Action Plan with measures potentially grouped into proposed stages for implementation according to payback - typically 0-1 year, 1-2 years, over 2 years.

It is intended that the final update report should reasonably ensure understanding of the principles of savings ideas and allow instruction on implementation of most basic improvements and changes to operational procedures. Detailed design of more complex solutions is generally excluded and is required to be provided later as a turn-key by your designated contractor or via a Building Services Engineering consultant. AEM can also assist you with this detailing stage as a separate independent consulting commission – for example in specifying installation details and control requirements for inverter drives, heat recovery etc., or in providing project management and commissioning supervision for improvement works.

We shall also identify in our report any situation that is observed where there is scope to improve maintenance, durability of plant and on a Duty of Care basis, wherever we observe a safety issue and we also include a brief formal review of main IT room resilience.

MAIN ASSUMPTIONS AND EXCLUSIONS

1. You will provide complimentary guest-standard rooms with broadband internet access for the duration of each site investigation. Meals shall be available as (1) breakfasts in guest accommodation (2) Lunch would be taken in staff canteen. (3) Option for either room service or guest restaurant evening meals (4) we shall be eligible for 50% discount on restaurant alcoholic beverages.

(Please note we would aim to sample multiple restaurants at least once to assess environmental experience at first hand from guest perspective.)

Staff / accommodation grade	Days of staying	Duties
Project In-charge	4 nights	Kick off meeting & system general checking
Project Engineer 1		Data & drawings checking
Project Engineer 2		Detail Equipment checking

2. All air fare costs will be the responsibility of the client.
3. We include infra-red thermal camera bring our basic test equipment to include portable ammeters and temperature / humidity / lux meters etc.
4. We shall require desk / base for 3 persons in your Engineering Department and 3 x bicycles whilst present on site.
5. We would anticipate presenting basic findings to senior staff at site as a briefing prior to departure and to follow with a full report typically 4-6 weeks thereafter.
6. At this stage we specifically exclude (1) detailed M&E Engineering Design Consultancy (2) NFPA 70B thermographic inspection services (3) Project Management services (4) supervision of Testing and Commissioning, but we should be pleased to quote for these services if required.
7. We shall use our best endeavours in the investigation of all savings opportunities reasonably apparent at the time of inspection but shall not be liable for any omission or ultimate saving delivery differing from our necessarily brief assessment, which must incorporate assumptions provided by your staff for the seasonal usage of plant.

8. In the event of any extra unplanned prolongation at site due to travel delay beyond our control (e.g. flight cancellation or adverse weather) we assume you will provide required extension of accommodation and meals provision at no cost. In this eventuality we would not apply any additional costs to you provided delays did not exceed 48 hours and we were granted continued access to plant areas to augment our investigations and allow us to commence report preparation at site. You will arrange at your cost all applicable transfers between airports and site / inter site flights and any applicable arising flight re-booking charges
9. Report to be provided in PDF format.

Travelling Schedule

Subject to final arrangement with Hotels

Final timetabling and on-duration of this audit to be confirmed for on-site checking plus report preparation time. The final date will be subject to confirmation

CHARGES

The lump sum fee for this investigation excluding air fares including site survey, travelling time and report preparation will be **USD \$9,925 net of tax** (rate per Four Seasons APAC group / AEM Agreement). You will be liable for meeting any required extra local taxes when dispatching the clear funds above at the conclusion of work, with **final net payment to us of the above sum. Payment will be due on completion to be defined by submission of final report as soft copy PDF file.**

Please sign and return the duplicate for your confirmation and acceptance.

Yours faithfully,
For and on behalf of
Active Energy Management Limited



Mike Wong
General Manager

Confirmed and accepted by

